FINS3616 International Business Finance

Term 1, 2025

Week 1 Reading Guide and Tutorial Questions

Chapters 1, 2, Multinational Corporations (MNCs) and Exchange Rate Determination

Week 1 Readings (Shapiro)

Chapter 1 – Introduction: Multinational Enterprise and Multinational Financial Management

1.1 The Rise of the Multinational Corporation

1.2 The Internationalization of Business and Finance

Chapter 2 – The Determination of Exchange Rates

2.1 Setting the Equilibrium Spot Exchange Rate

2.2 Expectations and the Asset Market Model of Exchange Rates - *only content covered in lecture*

2.3 The Fundamentals of Central Bank Intervention - *only content covered in lecture*

Week 1 Tutorial Questions

Chapter 1:

MNCs: Conceptual Question 2 and 7

Chapter 2:

Exchange Rate Changes: Conceptual Questions 1, 2, 4 and 5

Currency Appreciation/Depreciation: Problems 4 and 5

Additional questions on Currency Appreciation/Depreciation.

Chapter 1

Chapter 1, Question 2

2.a. How does foreign competition limit the prices that domestic companies can charge and the wages and benefits that workers can demand?

ANSWER. As domestic producers raise their prices, customers begin substituting less expensive goods and services supplied by foreign producers. The likelihood of losing sales limits the prices that domestic firms can charge. Foreign competition also acts to limit the wages and benefits that workers can demand. If workers demand more money, firms have two choices. Acquiesce in these demands or fight them. Absent foreign competition, the cost of acquiescence is relatively low, particularly if the industry is unionized. Since all firms will face the same higher costs, they can cover these higher costs by all of them simultaneously raising their prices without fear of being undercut or of being placed at a competitive disadvantage relative to their peers. Foreign competition changes the picture since foreign firms' costs will be unaffected by higher domestic wages and benefits. If domestic firms give in on wages and benefits, foreign firms will underprice them in the market and take market share away. In this case, higher domestic costs will put domestic firms at a disadvantage vis-à-vis their foreign competitors. Recognizing this, domestic firms facing foreign competition are more likely to fight worker demands for higher wages and benefits.

b. What political solutions can help companies and unions avoid the limitations imposed by foreign competition?

ANSWER. The classic political solution is protectionism. By limiting foreign competition, either through tariffs or quotas, companies and workers limit the ability of foreign goods to restrain domestic price increases. The government can also subsidize domestic firms in competing against foreign firms. These subsidies allow domestic firms and unions to perpetuate uneconomic work rules, wages, and productions processes.

c. Who pays for these political solutions? Explain.

ANSWER. Consumers pay for protectionism in the form of higher prices for their goods and service, fewer choices, and lower quality. These consumers include firms that use imports to produce their own goods and services for sale. Taxpayers pay for subsidies in the form of higher taxes or fewer of the other services provided by government

Chapter 1, Question 7

7. *Is there any reason to believe that MNCs may be less risky than purely domestic firms? Explain.*

ANSWER. Yes. International diversification may actually allow firms to reduce the total risk they face. Much of the general market risk facing a company is related to the cyclical nature of the domestic economy of the home country. Operating in a number of nations whose economic cycles are not perfectly in phase may, therefore, reduce the overall variability of the firm's earnings. Thus, even though the riskiness of operating in any one country may exceed the risk of operating in the United States (or other home country), much of that risk is eliminated through diversification. In fact, as shown in Chapter 15, the variability of earnings appears to decline as firms become more internationally oriented.

<u>Chapter 2, Question 1</u>

In February 2022, as a response to its invasion of Ukraine, Russia faced unprecedented sanctions from Western countries representing more than half the world's economic output. What is the expected market reaction for the Russian Ruble (RUB)?

ANSWER. According to Reuters, "The Russian currency tumbled to a record low 121.5275 in the weeks after Moscow's February 2022 invasion of Ukraine." This is to be expected since sanctions reduce the purchasing power of the sanctioned currency.

Chapter 2, Question 2

2.2 In March 2020, the COVID-19 pandemic causes almost all transportation to cease in major industrialized countries. What would be the likely market reaction for the Norwegian krone (NOK) and the Saudi Arabian riyal (SAR) given that these two countries are large petroleum exporters?

ANSWER. With the price of petroleum likely to drop, the value of currencies of petroleum exporters is likely to drop as well. For the NOK, the open price of the dollar against the Krone was 9.4130 on March 2, 2020. By December 30, 2020, it was down to 8.5292. For the riyal, on March 2, 2020, $1 USD = ﷼3.7522. For December 31, 2020, $1 USD = ﷼3.7512 so COVID-19 has much less impact on the riyal.

Chapter 2, Question 4

2. The maintenance of money's value is said to depend on the monetary authorities. What might the monetary authorities do to a currency that would cause its value to drop?

ANSWER. The value of any good or asset is driven by its scarcity. What the monetary authorities could do is to make money less scarce by issuing more of it. This would lower its scarcity value. Even though its nominal value will always be the same, the added supply will reduce the purchasing power per unit of money.

Chapter 2, Question 5

3. For each of the following six scenarios, say whether the value of the dollar will appreciate, depreciate, or remain the same relative to the Japanese yen. Explain each answer. Assume that exchange rates are free to vary and that other factors are held constant.

a. The growth rate of national income is higher in the United States than in Japan.

ANSWER. The value of the dollar should rise as more rapidly rising GNP in the United States leads to a relative increase in demand for dollars.

b. Inflation is higher in the U.S. than in Japan.

ANSWER. The value of the dollar should fall in line with purchasing power parity. *c.*

Prices in Japan and the United States are rising at the same rate.

ANSWER. The exchange rate should remain the same.

d. Real interest rates are higher in the United States than in Japan.

ANSWER. The value of the dollar should rise as the higher real rates attract capital from Japan that must first be converted into dollars.

e. *The United States imposes new restrictions on the ability of foreigners to buy American companies and real estate.*

ANSWER. The value of the dollar should fall as foreigners find it less attractive to own U.S. assets.

f. *U.S. wages rise relative to Japanese wages, and American productivity falls behind Japanese productivity.*

ANSWER. Higher U.S. wages and declining relative productivity weaken the American economy and make it less attractive for investment purposes. Assuming that a weak economy leads to a weak currency, the dollar will fall. From a somewhat different perspective, when a nation's productivity growth lags behind that of its major trading partners, the other countries will become more depreciating currency is the market's way of restoring balance. The lagging country regains its balance, but only by accepting a lower real price for its goods. In effect, the cheaper currency is the market's way of cutting wages in the lagging country.

Chapter 2, Problem 4

4. In early August 2002 (the exact date is a state secret), North Korea reduced the official value of the won from \$0.465 to \$0.0067. The black market value of the won at the time was \$0.005. a. By what percent did the won devalue?

ANSWER. Using Equation 2.1, the won devalued (\$0.067-\$0.465)/\$0.465)=98.56%

Following the initial devaluation what further percentage devaluation would be necessary for the won to equal its black market value?

ANSWER. 25.4%

Chapter 2, Problem 5

Please note that prior to the formation of the Euro as a single currency within the Euro-zone area, each country, like Germany (Deutch Mark-DM), Itay (Lira) and France (French Franc, FF), had their own currencies.

5. *On Friday, September 13, 1992, the Italian lira was worth DM 0.0013065. Over the weekend, the lira devalued against the DM to DM 0.0012613.*

a. *By how much has the lira devalued against the DM?*

ANSWER. Using Equation 2.1, the lira devalued by (0.0012613 - 0.0013065)/0.0013065, or -3.46%.

b. *By how much has the DM appreciated against the lira?*

ANSWER. Using Equation 2.2, the DM appreciated against the lira by [(1/0.0012613) - (1/0.0013065)]/(1/0.0013065), or 3.58%.

c. *Suppose Italy borrowed DM 4 billion, which it sold to prop up the lira. What were the Bank of Italy's lira losses on this currency intervention?*

ANSWER. Prior to devaluation, DM billion was worth Lit (4 billion/0.0013065). Following devaluation, the DM 4 billion borrowing would cost Lit (4 billion/0.0012613) to repay. Hence, the Italian government would lose Lit 4 billion x [(1/0.0012613) - (1/0.0013065)] = Lit 109,716,164344, or DM 138,384,998 at the new exchange rate.

d. *Suppose Germany spent DM 24 billion in an attempt to defend the lira. What were the Bundesbank's DM losses on this currency intervention?*

ANSWER. The Bundesbank would have bought Lit 24 billion/0.0013065. Following lira devaluation, these lira would be worth DM (24 billion/0.0013065) x 0.0012613, or DM 23,169,690,012. The result is a foreign exchange loss for the Bundesbank of DM 830,309,988 on this currency intervention.

Additional Question if time permits

1. Describe how these three typical transactions should affect present and future exchange rates.

a. Joseph E. Seagram & Sons imports a year's supply of French champagne. Payment in euros is due immediately.

ANSWER. The euro should appreciate relative to the dollar since demand for euros is rising.

b. MCI sells a new stock issue to Alcatel, the French telecommunications company. Payment in dollars is due immediately.

ANSWER. The spot value of the dollar should increase as Alcatel demands dollars to pay for the new stock issue. The future value of the dollar should decline as dividend payments are sent to Alcatel and other Alcatel equipment and parts are imported. However, the value of the dollar in the future could increase if expanded MCI output substitutes for telecom imports.

c. Korean Airlines buys five Boeing 747s. As part of the deal, Boeing arranges a loan to KAL for the purchase amount from the U.S. Export Import Bank. The loan is to be paid back over the next seven years with a two year grace period.

ANSWER. The spot price of the dollar should be unaffected. The future price of the dollar should increase as KAL repays the loan.

Additional Problems if time permits

1. On August 8, 2000, Zimbabwe changed the value of the Zim dollar from Z$38/U.S.$ to Z$50/U.S.$.

a. What was the original U.S. dollar value of the Zim dollar? What is the new U.S. dollar value of the Zim dollar?

ANSWER. The U.S. dollar value of the Zim dollar prior to devaluation was $0.0263 (1/38). Subsequent to devaluation, the Zim dollar was worth $0.02 (1/50).

b. By what percent has the Zim dollar devalued (revalued) relative to the U.S. dollar?

ANSWER. The U.S. dollar value of the Zim dollar has changed by (0.02 - 0.0263)/0.0263 = -24%. Thus, the Zim dollar has devalued by 24% against the U.S. dollar.

c. By what percent has the U.S. dollar appreciated (depreciated) relative to the Zim dollar?

ANSWER. The U.S. dollar has appreciated against the Zim dollar by an amount equal to (50 - 38)/38 = 31.58%.

2. In 1995, one dollar bought ¥80. In 2000, it bought about ¥110.

a. What was the dollar value of the yen in 1995? What was the yen's dollar value in 2000?

ANSWER. The dollar value of the yen in 1995 was \$0.0125 (1/80). By 2000, the yen had fallen to \$0.00909.

b. By what percent has the yen fallen in value between 1995 and 2000?

ANSWER. Between 1995 and 2000, the yen fell by 27.27%, calculated as (0.00909 - 0.0125)/0.0125.

c. By what percent has the dollar risen in value between 1995 and 2000?

ANSWER. During this same period, the dollar appreciated by 37.5%, calculated as (110 - 80)/80.

3. On February 1, the euro is worth \$0.8984. By May 1, it has moved to \$0.9457.

a. By how much has the euro appreciated or depreciated against the dollar over this 3-month period?

ANSWER. Since the euro is now worth more in dollar terms, it has appreciated against the dollar. The amount of euro appreciation is (0.9457 - 0.8984)/0.8984 = 5.27%.

b. By how much has the dollar appreciated or depreciated against the euro over this period?

ANSWER. The flip side of euro appreciation is dollar depreciation. The dollar has depreciated by an amount equal to

$$\frac{\frac{1}{0.9457} - \frac{1}{0.8984}}{\frac{1}{0.8984}} = \frac{0.8984 - 0.9457}{0.9457} = -5.00\%$$